

# Minh Quan (Andy) Quach

Phone | Email | LinkedIn | GitHub | Portfolio

## SUMMARY

Software Engineer with 5+ years of experience building production software and integrating backend systems across large-scale applications. Strong foundation in JavaScript, Python, SQL, REST APIs, and system integration, with graduate studies in Software Engineering at the University of Waterloo. Passionate about building scalable software, automating business workflows, and exploring AI-powered solutions to improve operational efficiency.

## TECHNICAL SKILLS

**Languages:** JavaScript, Python, SQL  
**Frontend:** React, Next.js, HTML, CSS  
**Backend:** REST APIs, Node.js, System Integration  
**Cloud:** AWS (S3, Lambda, API Gateway)  
**Databases:** PostgreSQL, MySQL  
**Tools:** Git, Linux, Docker (Basic), Jira, Confluence  
**AI Tools:** GitHub Copilot, ChatGPT, Claude

## PROFESSIONAL EXPERIENCE

**Software Engineer** 2015 – 2023  
**Elinext • Ho Chi Minh City, Vietnam**

- Developed and maintained large-scale mobile applications serving parking, mobility, and customer transaction services across multiple international markets
- Key Products:
  - **Flowbird** – leading parking and mobility platform used across Europe and North America
  - **HKeMeter** – official mobile application for parking meter payments in Hong Kong
- Designed, implemented, tested, and maintained frontend and backend features across multiple production applications.
- Integrated REST APIs and third-party services to support core application functionality
- Collaborated with cross-functional teams including product managers, QA engineers, and backend developers
- Debugged and improved application performance and user experience in production systems
- Participated in the full software development lifecycle including design, implementation, testing, deployment, and maintenance.

## PROJECTS

**2025-Present**

**Full-Stack Portfolio Platform (Next.js + AWS Serverless)** 2026

- Built a full-stack portfolio platform using Next.js, TypeScript, and Tailwind CSS with a modular and scalable architecture.
- Developed responsive, mobile-first user interfaces with reusable React components and dynamic routing.
- Designed and implemented RESTful APIs using AWS Lambda and API Gateway to serve dynamic portfolio content.
- Built a serverless backend integrated with Amazon DynamoDB for structured content and asset management.
- Implemented CI/CD using GitHub Actions to automate frontend and backend deployments.
- Designed an extensible admin system supporting authentication-based access control and CRUD operations for portfolio content management (projects, certifications, achievements, tech stack)
- Applied production-level software engineering best practices including modular design, version control, separate of concerns, and reusable component development.

## **Multi-Source Data Integration Pipeline (AWS S3 + Glue + Athena + Lambda)** 2026

- Built a serverless ETL pipeline on AWS to ingest transaction (CSV) and product API data into an S3 data storage
- Implemented API ingestion using AWS Lambda to fetch and store semi-structured JSON data in S3
- Transformed raw data into partitioned Parquet datasets using AWS Glue (PySpark)
- Modeled fact (fact\_transactions) and dimension (dim\_products) tables for analytics
- Developed analytical SQL queries in Athena to generate insights on revenue, top products, and category performance
- Implemented data validation and reconciliation checks during PySpark transformations to ensure analytics-ready datasets

## **ETL Log Pipeline (Python + PostgreSQL + Airflow)** 2025

- Built a backend service processing 1.8M+ records with API-style data ingestion and structured transformation logic.
- Implemented REST-style data ingestion logic with validation and error handling
- Designed database schema and optimized batch insert using PostgreSQL COPY
- Built modular Python services for parsing, cleaning, and transforming raw data

## **EDUCATION**

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### **Master of Engineering (MEng), Electrical & Computer Engineering** 2024

University of Waterloo — Waterloo, ON

Specialization: Software    Average: 87%

#### **Relevant Coursework:**

- ECE 650 – Methods and Tools for Software Engineering (92%)
- ECE 651 – Foundations of Software Engineering (89%)
- ECE 653 – Software Testing, Quality Assurance and Maintenance (83%)
- ECE 606 – Algorithm Design and Analysis (75%)

### **Bachelor of Computer Science** 2015

University of Science — Ho Chi Minh City    Average: 84%

## **CERTIFICATIONS**

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Databricks Certified Data Engineer Associate — Credential

IBM Data Engineering Professional Certificates — Coursera

AWS Certified Cloud Practitioner (CLF-C02) — Credential